

Isometric m -type edges and symmetries

Kaoru IINO

Yokohama National University

iino-kaoru-rb@ynu.jp

Abstract

We study surfaces with singularities. We classify isometric m -type edges up to right-equivalence and congruence.

§1 Introduction

f : two-parameter surface with singular points, ds_f^2 : the first fundamental form of f ,
 c : the singular curve of f , C : the image of c .

Fact 1 [1]

Q. If ds_f^2 and C are fixed, how many cuspidal edges have these data?

A. Under the admissible*, there exist **exactly 4 cuspidal edges** (* Roughly speaking, singular set is a curve).

The class of m -type edges was introduced in [2].

• Especially, we want to study the case m is odd.

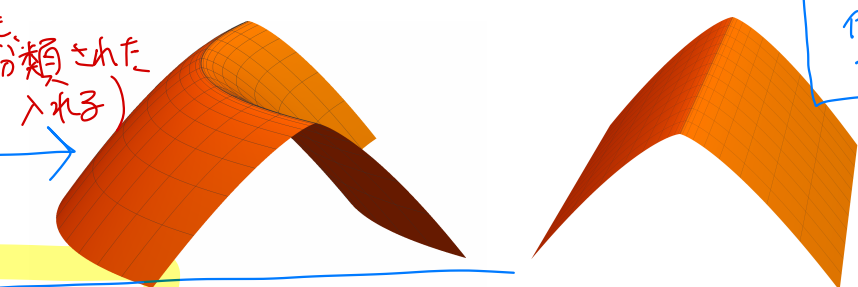


Figure 1: Admissible 3-type edge (left) and non-admissible 3-type edge (right).

§2 Right-Equivalence Class Count

Definition

A C^ω -map f has an m -type edge at $p \iff$ locally,
 $f(u, v) = (u, v^m, v^{m+1}\alpha(u, v))$ for some C^∞ function α .

For example, f is

- ▷ a cuspidal edge $\iff (m, \alpha) = (2, 1)$,
- ▷ a (3, 4)-cuspidal edge $\iff (m, \alpha) = (3, 1)$,
- ▷ a (3, 5)-cuspidal edge $\iff (m, \alpha) = (3, v)$.

Fact 2 [3]

f is an admissible m -type edge
 $\implies \exists! \check{f}, f_*, \check{f}_*$ s.t. having the same ds_f^2 and C .

Q. How many different surface shapes do they represent?

- ▷ f, g are **right-equivalence** $\iff \exists$ diffeo φ s.t. $f = g \circ \varphi$.
- ▷ φ is **symmetry** of ds_f^2 $\iff \varphi^* ds_f^2 = ds_f^2$ and $\varphi \neq \text{id}$.
- ▷ φ is **effective** (resp. **non-effective**) $\iff \varphi$ **reverses** (resp. **preserves**) the orientation of the singular curve.

[Main Thm 1]

Let f be an admissible m -type edge, and n_f be the number of right-equivalence classes of $\{f, \check{f}, f_*, \check{f}_*\}$. Then,

(1) (Generalization of the previous study [1]) If m is even,

- (a) $n_f = 2, 4$.
- (b) $n_f = 4 \iff ds_f^2$ has no effective symmetry.

(2) If m is odd,

- (a) $n_f = 1, 2, 4$.
- (b) $n_f = 4 \iff ds_f^2$ has **no symmetry**.
- (c) $n_f = 1 \iff ds_f^2$ has both **effective** symmetry and **non-effective** symmetry.

§3 Congruence Class Count

Next, we count different surface shapes **after rotations** in $\mathbb{R}^3$.

- ▷ f, g are **congruent** $\iff \exists T \in O(3)$ s.t. $f = Tg \circ \varphi$.
- ▷ T is **symmetry** of C $\iff T(C) = C$ and $T \neq \text{id}$.
- ▷ T is **C-orientation reversing** (resp. **preserving**) $\iff T$ **reverses** (resp. **preserves**) the orientation of C .

Remark $\exists C$ orientation preserving $\iff C$ is a **plane curve**.

[Main Thm 2]

Let f be an admissible m -type edge, and N_f be the number of congruence classes of $\{f, \check{f}, f_*, \check{f}_*\}$. Then $N_f \neq 3$. Moreover,

(1) (Generalization of the previous study [1]) If m is even,

- (I) $N_f = 4 \iff C$ has no symmetry and ds_f^2 has no effective symmetry.
- (II) $N_f = 1 \iff$ one of the following holds:
 - (a1) C is a plane curve and has a C-orientation-reversing symmetry.
 - (a2) C is a plane curve and ds_f^2 has an effective symmetry.
 - (b) C has a C-orientation-reversing symmetry with determinant 1, and ds_f^2 has an effective symmetry.

(2) If m is odd,

- (I) $N_f = 4 \iff$ **neither** ds_f^2 nor C has a **symmetry**.
- (II) $N_f = 1 \iff$ one of the following holds:

- (a1) C is a **plane curve** and has a C-orientation-reversing symmetry.
- (a2) C is a **plane curve** and ds_f^2 has an **effective** symmetry.
- (b1) ds_f^2 has a **non-effective** symmetry, and C has a C-orientation-reversing symmetry.
- (b2) ds_f^2 has both a **non-effective** symmetry and an **effective** symmetry.
- (c) C has a C-orientation-reversing symmetry with determinant 1, and ds_f^2 has an **effective** symmetry with negative Jacobian.
- (d) C has a C-orientation-reversing symmetry with determinant -1 , and ds_f^2 has an **effective** symmetry with positive Jacobian.

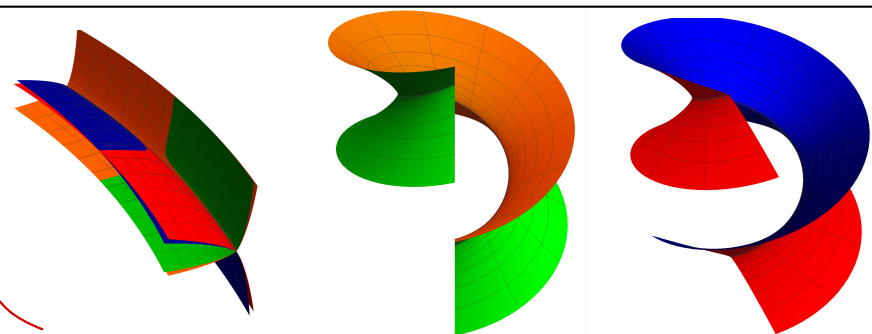


Figure 2: Examples of (3, 4)-cuspidal edges: an example satisfying $n_f = 4$ and $N_f = 1$

(left), and an example satisfying $n_f = 2$ and $N_f = 2$: $f, \check{f}, f_*$, and $\check{f}_*$.

For the latter example, the images of f and f_* (center), and $\check{f}$ and $\check{f}_*$ (right).

References

- [1] A. Honda, K. Naokawa, K. Saji, M. Umehara and K. Yamada, *Duality on generalized cuspidal edges preserving singular set images and first fundamental forms*, J. Singul. **22** (2020), 59–91.
- [2] L.F. Martins, K. Saji, S.P. dos Santos and K. Teramoto, *Boundedness of geometric invariants near a singularity which is a suspension of a singular curve*, Revista de la UMA **67** (2024), No. 2, 475–502.

[3] Y. Hattori, (3, 4)-カスプ辺の内在的な不変量, 修士論文, 横浜国立大学, 2025.